

Excel Lookup & Reference Functions

Complete Reference Guide

LOOKUP • VLOOKUP • HLOOKUP • XLOOKUP

INDEX • MATCH • INDEX+MATCH

6 Criteria · 7 Functions · Aryan Analytics Hub

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LOOKUP

The Classic — Simple One-Dimensional or Vector Lookup

1. Purpose

LOOKUP searches for a value in a one-row or one-column range and returns a value from the same position in a second range. It is the oldest of Excel's lookup family and works in two forms: **Vector Form** (search one row/column, return from another) and **Array Form** (search the first row or column of an array and return the last row or column). The lookup vector **must be sorted in ascending order**; otherwise results are unpredictable.

2. Syntax & Arguments

```
=LOOKUP(lookup_value, lookup_vector, [result_vector])
```

Argument	Required	Description
lookup_value	✓ Yes	The value you want to find. Can be a number, text, logical value, or a reference.
lookup_vector	✓ Yes	A one-row or one-column range that contains the search values. Must be sorted ascending.
[result_vector]	Optional	A one-row or one-column range of the same size as lookup_vector. If omitted, lookup_vector is used as the return range too (Array Form).

3. Business Use Cases

- Assigning grade letters (A, B, C) based on score thresholds stored in a sorted table.
- Translating product codes to descriptions when the code list is pre-sorted.
- Returning the last non-empty value in a column using the trick: =LOOKUP(2, 1/(A:A<>""), A:A).
- Simple price tier lookups where price bands are fixed and sorted.

4. Limitations & Common Mistakes

- Lookup vector **MUST** be sorted ascending — unsorted data returns wrong results silently.
- Only approximate match; no exact-match mode like VLOOKUP's FALSE argument.
- Cannot look right-to-left — the result column must be to the right of the search column.
- No error handling built-in; wrap with IFERROR for clean outputs.
- Largely superseded by XLOOKUP in modern Excel — avoid for new work.
- **Common mistake:** Forgetting to sort the lookup vector, leading to wrong values returned without any error.

5. Comparison with Other Lookup Functions

LOOKUP is the least flexible of the family. Unlike VLOOKUP, it isn't constrained to a fixed column number, and the search and result ranges can be completely separate — but it *requires sorted data* and only does approximate matching. XLOOKUP replaces it entirely: exact + approximate, unsorted data, error handling, and bidirectional lookup.

6. Edge Cases

- If lookup_value is smaller than the smallest value in the vector, LOOKUP returns #N/A.
- If lookup_value is larger than all values, LOOKUP returns the last value (approximate match behaviour).
- The 'last non-empty' trick works because 1/("") produces a #DIV/0 error which LOOKUP skips, landing on the last valid value.
- Array Form: if the array has more rows than columns, LOOKUP searches the first column; otherwise the first row.

VLOOKUP

The Workhorse — Vertical Lookup by Column Number

1. Purpose

VLOOKUP (Vertical Lookup) searches for a value in the **leftmost column** of a table and returns a value from a specified column in the same row. It is the most widely used lookup function in business Excel and works in two modes: **Exact Match** (range_lookup = FALSE / 0) and **Approximate Match** (range_lookup = TRUE / 1, requires sorted data). The function always searches left-to-right — the lookup column must be the first column of your table_array.

2. Syntax & Arguments

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

Argument	Required	Description
lookup_value	✓ Yes	The value to search for in the first column of the table. Can be a value, cell reference, or formula result.
table_array	✓ Yes	The range containing both the lookup column (leftmost) and the return column. Lock with \$ for reuse.
col_index_num	✓ Yes	Column number within table_array to return (1 = first column, 2 = second, etc.). Hard-coding this number is a common fragility.
[range_lookup]	Optional	TRUE or 1 = approximate match (sorted data required). FALSE or 0 = exact match. Always use FALSE in business contexts unless doing tiered lookups.

3. Business Use Cases

- Pulling employee salary from an HR table using Employee ID as the key.
- Matching invoice numbers to payment amounts in accounts payable reconciliation.
- Tax bracket lookups: approximate match on sorted income thresholds to find the applicable rate.
- Product catalogue lookups: fetch price, category, or supplier from a master table using SKU.
- Joining two datasets in Excel before Power Query is available.

4. Limitations & Common Mistakes

- Lookup column must be the leftmost column — cannot search right-to-left.
- col_index_num is positional and fragile: inserting/deleting columns silently breaks the formula.
- Returns the first match only; duplicate lookup values cause incorrect results.
- Approximate match (TRUE) requires ascending sort — unsorted data returns wrong values without error.

- Performance degrades on very large datasets vs INDEX+MATCH (scans entire table_array).
- **Common mistake #1:** Omitting range_lookup (defaults to TRUE = approximate match), returning wrong values on unsorted data.
- **Common mistake #2:** Hard-coding col_index_num (e.g. 3) — inserting a column shifts the data but not the index.
- **Common mistake #3:** Not locking table_array with \$ when copying the formula down.

5. Comparison with Other Lookup Functions

VLOOKUP is more constrained than INDEX+MATCH (leftmost-column limitation, fragile column index) and less powerful than XLOOKUP (which is bidirectional, handles errors natively, and supports wildcard and exact modes simultaneously). HLOOKUP is the horizontal equivalent. For new work, prefer XLOOKUP or INDEX+MATCH.

6. Edge Cases

- Partial/wildcard match: =VLOOKUP("Smith*", A:C, 2, FALSE) matches the first cell starting with 'Smith'.
- Case-insensitive: VLOOKUP treats 'apple' and 'APPLE' as identical — use EXACT() for case-sensitive lookups.
- Multiple matches: only the first match in the lookup column is returned; all subsequent matches are ignored.
- If col_index_num exceeds the number of columns in table_array, Excel returns #REF!.
- A col_index_num of 0 or negative returns #VALUE!.
- lookup_value longer than 255 characters returns #VALUE! — a hidden gotcha with long text keys.

HLOOKUP

Horizontal Lookup — Search Across the Top Row

1. Purpose

HLOOKUP (Horizontal Lookup) is the horizontal mirror image of VLOOKUP. It searches for a value in the **top row** of a table and returns a value from a specified row in the same column. Use HLOOKUP when your data is organised horizontally — headers across the top, data flowing downward. The same exact/approximate match modes apply as in VLOOKUP.

2. Syntax & Arguments

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

Argument	Required	Description
lookup_value	✓ Yes	The value to find in the first (top) row of the table.
table_array	✓ Yes	The horizontal range. The lookup row must be the topmost row.
row_index_num	✓ Yes	Row number within table_array to return (1 = top row, 2 = second row, etc.).
[range_lookup]	Optional	TRUE = approximate match (top row must be sorted). FALSE = exact match. Always use FALSE for most business scenarios.

3. Business Use Cases

- Retrieving monthly budget figures where months are column headers across the top.
- Pulling product attributes from a horizontally-laid-out product matrix.
- Score band lookup: months across the top, metrics down the side.
- Configuration tables where settings are named columns and values are in rows below.

4. Limitations & Common Mistakes

- Lookup row must be the topmost row — mirrors VLOOKUP's leftmost-column restriction.
- row_index_num is positional and fragile: inserting rows above the return row breaks the formula silently.
- Rarely used because most business data is organised vertically (columns), making VLOOKUP more applicable.
- Returns only the first match.
- **Common mistake:** Using HLOOKUP on a transposed dataset when VLOOKUP or INDEX+MATCH on the original orientation would be simpler.
- No native error handling — always wrap with IFERROR.

5. Comparison with Other Lookup Functions

HLOOKUP is the horizontal counterpart of VLOOKUP. In practice it is far less common because business data is predominantly vertical. INDEX+MATCH handles both orientations transparently, and XLOOKUP supports both row and column searches making both VLOOKUP and HLOOKUP redundant in modern Excel.

6. Edge Cases

- Wildcard matching works identically to VLOOKUP: ? matches any single character, * matches any sequence.
- If row_index_num = 1, HLOOKUP returns the lookup value itself (first row).
- If row_index_num exceeds the number of rows in table_array, Excel returns #REF!.
- Case-insensitive, same as VLOOKUP.
- lookup_value > 255 characters returns #VALUE!.

XLOOKUP

The Modern Standard — Flexible, Powerful, Error-Aware

1. Purpose

XLOOKUP is the modern replacement for VLOOKUP, HLOOKUP, and LOOKUP, introduced in Excel 365 / Excel 2021. It searches a lookup array and returns a corresponding value from a return array. Key advances: **bidirectional** (left, right, up, down), **built-in error handling** via `if_not_found`, **multiple match modes** (exact, approximate, wildcard), and **search direction** control (first-to-last or last-to-first). The lookup array and return array are *separate*, removing the `col_index` fragility of VLOOKUP.

2. Syntax & Arguments

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])
```

Argument	Required	Description
<code>lookup_value</code>	✓ Yes	The value to search for.
<code>lookup_array</code>	✓ Yes	The single row or column to search in. Completely independent of <code>return_array</code> .
<code>return_array</code>	✓ Yes	The single row or column to return from. Can be to the left of <code>lookup_array</code> .
<code>[if_not_found]</code>	Optional	Value to return when no match is found (replaces IFERROR wrapper). E.g. "Not Found" or 0.
<code>[match_mode]</code>	Optional	0 = exact (default), -1 = exact or next smaller, 1 = exact or next larger, 2 = wildcard.
<code>[search_mode]</code>	Optional	1 = first-to-last (default), -1 = last-to-first, 2 = binary ascending, -2 = binary descending.

3. Business Use Cases

- Left lookup: returning customer name (column A) based on invoice ID in column D — impossible with VLOOKUP.
- Last transaction lookup: `search_mode = -1` to find the most recent record for a customer.
- Approximate pricing tiers: `match_mode = -1` with unsorted tiers is not required to be sorted.
- Wildcard search: `match_mode = 2` allows partial string matching without helper columns.
- Error-safe dashboards: `if_not_found` returns a clean fallback instead of #N/A.
- Two-way lookup combined with MATCH to return a cell at a row/column intersection.

4. Limitations & Common Mistakes

- Only available in Excel 365, Excel 2021, and Excel Online — not in Excel 2019 or earlier.
- Cannot return multiple columns in a single formula without spilling (dynamic arrays required).
- **Common mistake:** Mixing up `lookup_array` and `return_array` order (especially when migrating from `VLOOKUP`).
- **Common mistake:** Using `match_mode = 2` (wildcard) unintentionally when the lookup value contains `*` or `?`.
- Binary search modes (2, -2) require sorted data — using them on unsorted data gives wrong results silently.

5. Comparison with Other Lookup Functions

`XLOOKUP` fully supersedes `VLOOKUP`, `HLOOKUP`, and `LOOKUP`. Compared to `INDEX+MATCH`, `XLOOKUP` is simpler to write and read (one function vs two nested), has native `if_not_found` error handling, and supports last-match and wildcard modes directly. `INDEX+MATCH` still has an edge for compatibility with older Excel versions and for some advanced multi-criteria scenarios.

6. Edge Cases

- `XLOOKUP` can return an entire row or column if `return_array` is multi-row/column — it spills the results (dynamic array).
- Nested `XLOOKUP` for two-way lookup: `=XLOOKUP(row_val, row_headers, XLOOKUP(col_val, col_headers, data_body))`.
- `if_not_found` is evaluated lazily — it only runs when no match exists, so it can contain another formula.
- `search_mode = -1` returns the LAST match, useful for finding the most recent record in unsorted append-only logs.
- If `lookup_array` and `return_array` sizes don't match, Excel returns `#VALUE!`.

INDEX

The Coordinate Retriever — Return a Value by Position

1. Purpose

INDEX returns the value at a given row and column position within a range or array. Think of it as a coordinate system: you give it a grid and a cell address (row number, column number), and it hands back the value at that intersection. On its own, INDEX is useful for referencing dynamic positions. Its true power emerges when combined with MATCH, which provides the row/column numbers dynamically.

2. Syntax & Arguments

`=INDEX(array, row_num, [col_num])` –or– `=INDEX(reference, row_num, [col_num], [area_num])`

Argument	Required	Description
array	✓ Yes	The range or array to return a value from.
row_num	✓ Yes	The row number within the array (1-based). Use 0 to return the entire column.
[col_num]	Optional	The column number within the array (1-based). Use 0 to return the entire row. If omitted, row_num selects from a single-column array.
[area_num]	Optional	When reference is a multi-area range (e.g. (A1:B5, D1:E5)), area_num selects which area to use. Default is 1.

3. Business Use Cases

- Dynamic range names: INDEX can return a reference used as the endpoint of a range — `=A1:INDEX(A:A, COUNTA(A:A))`.
- Retrieving values from the nth row/column of a filtered or sorted array.
- Combined with MATCH for flexible two-way table lookups.
- Returning entire rows or columns (row_num or col_num = 0) for use in other functions.
- Building dynamic chart ranges that expand automatically as data grows.

4. Limitations & Common Mistakes

- Requires knowing the row/column position — not directly useful for value-based searching without MATCH.
- Array Form vs Reference Form can be confusing; Reference Form requires non-contiguous ranges wrapped in parentheses.
- **Common mistake:** Using 0-based indexing — INDEX is 1-based, so row 1 is the first data row.
- **Common mistake:** Confusing `INDEX(array, 0, col)` to mean 'return whole column' vs using it in an array formula context.

5. Comparison with Other Lookup Functions

INDEX alone is a position-based retrieval tool, not a search function. It becomes a lookup tool only when paired with MATCH. Compared to VLOOKUP, INDEX+MATCH has no leftmost-column restriction and the return position is dynamic (not a hard-coded number). XLOOKUP is easier to write for most lookups, but INDEX retains advantages in dynamic range construction and certain array formula patterns.

6. Edge Cases

- INDEX(array, 0, col_num) returns the entire column as an array — useful as input to SUM, MATCH, etc.
- INDEX(array, row_num, 0) returns the entire row as an array.
- When row_num or col_num is 0 in a non-array context, Excel may return only the first value — enter as Ctrl+Shift+Enter (CSE) in older Excel.
- INDEX returns a reference (not just a value), so it can be used inside INDIRECT, OFFSET, or as a range argument.
- Negative row/column numbers return #VALUE!.

MATCH

The Position Finder — Where Is This Value?

1. Purpose

MATCH returns the **relative position** (a number) of a `lookup_value` within a one-dimensional array. It does not return the value itself — it returns the row or column number where the value was found. This makes it the perfect partner for INDEX: MATCH finds the position, INDEX uses that position to retrieve the value. MATCH supports three match types: exact (0), less-than-or-equal approximate (-1, sorted descending), and greater-than-or-equal approximate (1, sorted ascending).

2. Syntax & Arguments

```
=MATCH(lookup_value, lookup_array, [match_type])
```

Argument	Required	Description
<code>lookup_value</code>	✓ Yes	The value whose position you want to find. Can be text, number, logical, or wildcard.
<code>lookup_array</code>	✓ Yes	A single row or column range to search in.
<code>[match_type]</code>	Optional	0 = exact match (most common in business). 1 = largest value ≤ <code>lookup_value</code> (sorted ascending). -1 = smallest value ≥ <code>lookup_value</code> (sorted descending). Default is 1.

3. Business Use Cases

- Finding the column number of a header dynamically so INDEX+MATCH can reference it without hard-coding.
- Checking whether a value exists in a list — if MATCH returns #N/A, the value is absent.
- Returning the position of the maximum or minimum value: MATCH(MAX(range), range, 0).
- Building dynamic dropdown-driven reports where the selected header drives the column retrieved.
- Data validation: confirm a lookup key exists before performing further calculations.

4. Limitations & Common Mistakes

- Returns only the position, not the value — always needs INDEX (or equivalent) to be useful in most contexts.
- Exact match (0) requires no sorting; approximate match (1 or -1) requires sorted data.
- Returns only the first match position for duplicates.
- **Common mistake:** Omitting `match_type` — default is 1 (approximate, ascending sort required), which silently returns wrong results on unsorted data.
- **Common mistake:** Forgetting that MATCH is 1-based, so position 1 = the first item in the array.

5. Comparison with Other Lookup Functions

MATCH is a utility function, not a standalone lookup. It is the position-finding half of the INDEX+MATCH pair. XMATCH (Excel 365) is the modern equivalent with additional modes: wildcard (2), binary ascending (3), binary descending (-3), and a search_mode argument for last-match. For compatibility with Excel 2019 and earlier, MATCH remains the standard choice.

6. Edge Cases

- MATCH with wildcards (match_type = 0): =MATCH("A*", A1:A100, 0) finds the first cell starting with 'A'.
- MATCH(MAX(range), range, 0) returns the position of the largest value.
- If the array is sorted descending, use match_type = -1 to find the smallest value $\geq$ lookup_value.
- MATCH is case-insensitive for text; use an array formula with EXACT() for case-sensitive position finding.
- If lookup_value is not found and match_type = 0, MATCH returns #N/A — use IFERROR or IFNA to handle cleanly.

INDEX + MATCH

The Power Combination — Dynamic, Flexible, Bidirectional

1. Purpose

INDEX+MATCH is not a built-in Excel function but a **formula pattern** that combines the two functions to create a lookup more powerful than VLOOKUP. MATCH finds the row (and/or column) position of a lookup value; INDEX uses that position to retrieve the result. The combination eliminates VLOOKUP's key weaknesses: the lookup column can be *anywhere*, the return column reference is *dynamic* (not a hard-coded number), and it works in both vertical and horizontal directions. It remains the gold standard for Excel 2019 and earlier, and is still preferred by many professionals even in Excel 365.

2. Syntax & Arguments

```
=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))
```

Argument	Required	Description
return_range	✓ Yes	The column (or row) from which you want to retrieve a value.
lookup_value	✓ Yes	The value to search for.
lookup_range	✓ Yes	The column (or row) to search in — can be left of, right of, or same column as return_range.
0 (match_type)	✓ Yes	Use 0 for exact match in virtually all business cases.

3. Business Use Cases

- Left lookup: return values in column A (Name) by searching column C (ID) — impossible with VLOOKUP.
- Two-way lookup: =INDEX(data, MATCH(row_val, row_headers,0), MATCH(col_val, col_headers,0)) — retrieve a value at the intersection of a row and column.
- Dynamic column reference: MATCH(header_name, header_row, 0) generates the column number automatically — survives column insertions.
- Closest value lookup: MATCH(MIN(ABS(range - target)), ABS(range - target), 0) finds the nearest match.
- Multi-criteria lookup (array formula): =INDEX(C:C, MATCH(1, (A:A=x)*(B:B=y), 0)) — match on two conditions simultaneously.

4. Limitations & Common Mistakes

- Slightly more complex to write and read than XLOOKUP for simple lookups.
- Multi-criteria version requires Ctrl+Shift+Enter (CSE) in Excel 2019 or earlier (not needed in Excel 365 with dynamic arrays).
- Two separate functions to maintain — formula is longer than XLOOKUP equivalent.

- No native if_not_found argument — must wrap the entire formula in IFERROR.
- **Common mistake:** Mismatched ranges — INDEX return_range and MATCH lookup_range must have the same number of rows (or columns) and be aligned.
- **Common mistake:** Forgetting match_type = 0, letting it default to 1 (approximate, requires sorted data).

5. Comparison with Other Lookup Functions

INDEX+MATCH vs VLOOKUP: superior in every way — bidirectional, dynamic column reference, no leftmost-column restriction, better performance on large datasets (INDEX+MATCH only scans the lookup column, not the whole table_array). vs XLOOKUP: XLOOKUP is simpler, has native error handling, supports last-match and wildcard modes, but requires Excel 365/2021. INDEX+MATCH is the best choice for backward-compatible workbooks and for advanced array formula scenarios that XLOOKUP doesn't handle as cleanly.

6. Edge Cases

- Two-way lookup pattern: =INDEX(B2:E10, MATCH("Product A", A2:A10, 0), MATCH("Q3", B1:E1, 0)).
- Multi-criteria (CSE): {=INDEX(C2:C100, MATCH(1, (A2:A100=x)*(B2:B100=y), 0))} — enter with Ctrl+Shift+Enter in Excel 2019.
- In Excel 365, multi-criteria works without CSE: =INDEX(C2:C100, MATCH(x&y, A2:A100&B2:B100, 0)) — concatenate lookup keys.
- Nearest-match lookup: =INDEX(B:B, MATCH(MIN(ABS(A:A - target)), ABS(A:A - target), 0)).
- If the lookup range has no match and IFERROR is omitted, the whole formula returns #N/A — always wrap with IFERROR for production dashboards.

Quick Comparison Table

All 7 Functions at a Glance

Function	Direction	Sorted?	Left Lookup	Error Handling	Wildcard	Excel Version	Recommendation
LOOKUP	Both	Required	No	IFERROR wrap	No	All	Legacy only
VLOOKUP	Down	For approx	No	IFERROR wrap	Yes	All	Avoid new work
HLOOKUP	Right	For approx	N/A	IFERROR wrap	Yes	All	Rare use
XLOOKUP	Both	Optional	Yes	if_not_found arg	Yes	365 / 2021	Preferred
INDEX	Both	No	Yes	IFERROR wrap	No	All	Use with MATCH
MATCH	Both	For approx	Yes	IFERROR wrap	Yes	All	Use with INDEX
INDEX + MATCH	Both	No	Yes	IFERROR wrap	Yes	All	Best compatible

■ Rule of thumb: Use XLOOKUP if on Excel 365/2021. Use INDEX+MATCH for backward-compatible workbooks or multi-criteria lookups. Avoid VLOOKUP and HLOOKUP in new work.

■ Always wrap lookup formulas in IFERROR or use XLOOKUP's if_not_found argument in production dashboards — a #N/A in one cell can cascade errors through dependent formulas.